Huy Nguyen

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CONTACT INFORMATION

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EDUCATION

The University of Texas at Austin, Austin, TX, USA. 2022-2027
Ph.D. Candidate in Statistics at the Department of Statistics and Data Science.

- Advisors: Professor Nhat Ho and Professor Alessandro Rinaldo.
- GPA: 4.0/4.0

Ho Chi Minh University of Science, Ho Chi Minh City, Vietnam. 2017–2020
Bachelor of Science in Mathematics and Computer Science, Honor Program.

- Advisor: Professor Dang Duc Trong.
- GPA: 9.75/10.0 - *Summa Cum Laude*.

RESEARCH EXPERIENCE

Morgan Stanley, New York, NY, USA. Summer 2026
Summer Associate.

- Research topics: Mixture of Experts in Low-rank Adaptation for Language Models.

Microsoft Corporation, Redmond, WA, USA. Summer 2024
Research Intern.

- Research topics: Applications of Mixture of Experts in Language Models.

VinAI Research (acquired by Qualcomm AI Research), Hanoi, Vietnam. 2020–2022
AI Research Resident.

- Research topics: Optimal Transport theory and its applications in Domain Adaptation.

RESEARCH INTERESTS

My research broadly lies at the intersection of statistical machine learning and theoretical foundation for AI, with a particular focus on four important aspects of Mixture-of-Experts (MoE) models, including Foundation (statistical understandings of gating mechanism and expert estimation), Scalability (effective sparse MoE in language models), Heterogeneity (MoE in multi-modal learning), and Efficiency (MoE in parameter-efficient fine-tuning, namely low-rank adaptation and prompt-based tuning). Previously, I also worked on computational and approximation aspects of optimal transport.

PUBLICATIONS

27. Viet Nguyen*, Tuan Minh Pham*, Thinh Cao*, Tan Dinh, **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. A Statistical Theory of Gated Attention through the Lens of Hierarchical Mixture of Experts. *Advances in Neural Information Processing Systems, 2026.*
26. Minh Le, Bao-Ngoc Dao, **Huy Nguyen**, Quyen Tran, Anh Nguyen, Nhat Ho. One-Prompt Strikes Back: Sparse Mixture of Experts for Prompt-based Continual Learning. *In International Conference on Learning Representations, 2026.*
25. Minh Le, Anh Nguyen, **Huy Nguyen**, Chau Nguyen, Anh Tran, Nhat Ho. Revisit Visual Prompt Tuning: The Expressiveness of Prompt Experts. *In International Conference on Learning Representations, 2026.*
24. **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. Convergence Rates for Softmax Gating Mixture of Experts. *IEEE Transactions on Information Theory 72(2), 1276-1304, 2026.*
23. **Huy Nguyen**, Pedram Akbarian*, Trang Pham*, Trang Nguyen*, Shujian Zhang, Nhat Ho. Statistical Advantages of Perturbing Cosine Router in Mixture of Experts. *In International Conference on Learning Representations, 2025.*
22. Nghiem Tuong Diep*, **Huy Nguyen***, Chau Nguyen, Minh Le, Duy Minh Ho Nguyen, Daniel Sonntag, Mathias Niepert, Nhat Ho. On Zero-Initialized Attention: Optimal Prompt and Gating Factor Estimation. *Proceedings of the International Conference on Machine Learning, 2025.*

21. Tuan Truong*, Chau Nguyen*, **Huy Nguyen***, Minh Le, Trung Le, Nhat Ho. RepLoRA: Reparameterizing Low-rank Adaptation via the Perspective of Mixture of Experts. *Proceedings of the International Conference on Machine Learning, 2025.*

20. Minh Le*, Chau Nguyen*, **Huy Nguyen***, Quyen Tran, Trung Le, Nhat Ho. Revisiting Prefix-tuning: Statistical Benefits of Reparameterization among Prompts. *In International Conference on Learning Representations, 2025.*

19. Fanqi Yan*, **Huy Nguyen***, Dung Le*, Pedram Akbarian, Nhat Ho**, Alessandro Rinaldo**. On Minimax Estimation of Parameters in Softmax-Contaminated Mixture of Experts. *Advances in Neural Information Processing Systems, 2025..*

18. Fanqi Yan*, **Huy Nguyen***, Dung Le*, Pedram Akbarian, Nhat Ho. Understanding Expert Structures on Minimax Parameter Estimation in Contaminated Mixture of Experts. *In International Conference on Artificial Intelligence and Statistics, 2025.*

17. **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. Sigmoid Gating is More Sample Efficient than Softmax Gating in Mixture of Experts. *Advances in Neural Information Processing Systems, 2024.*

16. Xing Han, **Huy Nguyen***, Carl Harris*, Nhat Ho, Suchi Saria. FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion. *Advances in Neural Information Processing Systems, 2024.*

15. Minh Le, An Nguyen*, **Huy Nguyen***, Trang Nguyen*, Trang Pham*, Linh Van Ngo, Nhat Ho. Mixture of Experts Meets Prompt-Based Continual Learning. *Advances in Neural Information Processing Systems, 2024.*

14. **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. On Least Square Estimation in Softmax Gating Mixture of Experts. *Proceedings of the International Conference on Machine Learning, 2024.*

13. **Huy Nguyen**, Pedram Akbarian, Nhat Ho. Is Temperature Sample Efficient for Softmax Gaussian Mixture of Experts? *Proceedings of the International Conference on Machine Learning, 2024.*

12. **Huy Nguyen**, Pedram Akbarian, TrungTin Nguyen, Nhat Ho. A General Theory for Softmax Gating Multinomial Logistic Mixture of Experts. *Proceedings of the International Conference on Machine Learning, 2024.*

11. **Huy Nguyen**, Pedram Akbarian, Fanqi Yan, Nhat Ho. Statistical Perspective of Top-K Sparse Softmax Gating Mixture of Experts. *In International Conference on Learning Representations, 2024.*

10. **Huy Nguyen***, TrungTin Nguyen*, Khai Nguyen, Nhat Ho. Towards Convergence Rates for Parameter Estimation in Gaussian-gated Mixture of Experts. *In International Conference on Artificial Intelligence and Statistics, 2024.*

9. **Huy Nguyen**, Khai Nguyen, Nhat Ho. On Parameter Estimation in Gaussian Deviated Mixture of Experts. *In International Conference on Artificial Intelligence and Statistics, 2024.*

8. **Huy Nguyen**, TrungTin Nguyen, Nhat Ho. Demystifying Softmax Gating Function in Gaussian Mixture of Experts. *Advances in Neural Information Processing Systems, 2023 (Spotlight, Top 3.6% out of 12343 submissions).*

7. Dat Do*, **Huy Nguyen***, Khai Nguyen, Nhat Ho. Minimax Optimal Rate for Parameter Estimation in Multivariate Deviated Models. *Advances in Neural Information Processing Systems, 2023.*

6. Dung Le*, **Huy Nguyen***, Khai Nguyen*, Trang Nguyen*, Nhat Ho. Fast Approximation of the Generalized Sliced-Wasserstein Distance. *IEEE International Conference on Acoustics, Speech and Signal Processing, 2024.*

5. Khai Nguyen, Tongzheng Ren, **Huy Nguyen**, Litu Rout, Tan Nguyen, Nhat Ho. Hierarchical Sliced Wasserstein Distance. *In International Conference on Learning Representations, 2023.*

4. Khang Le*, Dung Le*, **Huy Nguyen***, Dat Do, Tung Pham, Nhat Ho. Entropic Gromov-Wasserstein between Gaussian Distributions. *Proceedings of the International Conference on Machine Learning, 2022.*

3. Khang Le*, **Huy Nguyen***, Khai Nguyen, Tung Pham, Nhat Ho. On Multimarginal Partial Optimal Transport: Equivalent Forms and Computational Complexity. *In International Conference on Artificial Intelligence and Statistics, 2022.*

2. Khang Le*, **Huy Nguyen***, Quang Minh Nguyen, Tung Pham, Hung Bui, Nhat Ho. On Robust Optimal Transport: Computational Complexity and Barycenter Computation. *Advances in Neural Information Processing Systems, 2021.*

1. Thu Nguyen, Duy H. M. Nguyen, **Huy Nguyen**, Binh T. Nguyen, Bruce A. Wade. EPEM: Efficient Parameter Estimation for Multiple Class Monotone Missing Data. *Information Sciences Journal, Volume 567, page 1-22.*

Preprints

14. **Huy Nguyen***, Dung Le*, Alessandro Rinaldo, Nhat Ho. On the Geometry of Separation in Finite Gaussian Mixtures. *Under review, arXiv:2606.16179.*

13. Dung Le*, **Huy Nguyen***, Trang Pham, Alessandro Rinaldo, Nhat Ho. Characterizing Heterogeneous Rates in Finite Mixture Estimation via Partial Optimal Transport. *Under review, arXiv:2609.16622.*

12. Sagar Ghosh, **Huy Nguyen**, Alessandro Rinaldo. Sparsity Saves Compute, Not Samples: Oracle Inequalities for Dense and Sparse Mixture of Experts . *Under review arXiv:2610.15039.*

11. Abbas Khalili*, Pengqi Liu*, **Huy Nguyen***, Nhat Ho. Joint Estimation and Feature Selection in Overspecified Finite Mixture of Regressions. *Under review.*

10. Nicola Barilleto*, **Huy Nguyen***, Nhat Ho**, Alessandro Rinaldo**. On Bayesian Softmax-Gated Mixture-of-Experts Models. *Under review, arXiv:2604.20551.*

9. Tuan Minh Pham*, Thinh Cao*, Viet Nguyen*, **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. Rethinking Multinomial Logistic Mixture of Experts with Sigmoid Gating Function. *Under review, arXiv:2602.01466.*

8. Fanqi Yan*, Dung Le*, Trang Pham, **Huy Nguyen**, Nhat Ho. Improving Minimax Estimation Rates for Contaminated Mixture of Multinomial Logistic Experts via Expert Heterogeneity. *Under review, arXiv:2602.00939.*

7. **Huy Nguyen**, Thong T. Doan, Quang Pham, Nghi Bui, Nhat Ho**, Alessandro Rinaldo**. On DeepSeekMoE: Statistical Benefits of Shared Experts and Normalized Sigmoid Gating. *Under review, arXiv:2505.10860.*

6. Nghiem T. Diep, Dung Le, Tuan Truong, Tan Dinh, **Huy Nguyen**, Nhat Ho. Hypernetwork-Driven Low-Rank Adaptation Across Attention Heads. *Under review, arXiv:2510.04295.*

5. Nghiem T. Diep, Hien Dang, Tuan Truong, Tan Dinh, **Huy Nguyen**, Nhat Ho. DoRAN: Stabilizing Weight-Decomposed Low-Rank Adaptation via Noise Injection and Auxiliary Networks. *Under review, arXiv:2510.04331.*

4. **Huy Nguyen***, Fanqi Yan*, Pedram Akbarian, Nhat Ho**, Alessandro Rinaldo**. Sigmoid Self-Attention has Lower Sample Complexity than Softmax Self-Attention: A Mixture-of-Experts Perspective. *Under review, arXiv:2502.00281.*

3. **Huy Nguyen***, Xing Han*, Carl Harris, Suchi Saria**, Nhat Ho**. On Expert Estimation in Hierarchical Mixture of Experts: Beyond Softmax Gating Functions. *Under review, arXiv:2410.02935.*

2. Pedram Akbarian*, **Huy Nguyen***, Xing Han*, Nhat Ho. Quadratic Gating Mixture of Experts: Statistical Insights into Self-Attention. *Under review, arXiv:2410.11222.*

1. Nam V. Nguyen, **Huy Nguyen**, Quang Pham, Van Nguyen, Savitha Ramasamy, Nhat Ho. CompeteSMoE - Statistically Guaranteed Mixture of Experts Training via Competition. *Under review, arXiv:2505.13380.*

Books

1. BayesAI Consortium. Handbook of Bayesian Deep Learning. *To be produced by CRC Press.*

Talks

7. Mixture of Experts in Large-scale and Multimodal Models. *Morgan Stanley Research Talk, New York, 2026 (Invited talk).*
6. FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion. *Jane Street Graduate Research Fellowship Workshop, New York, 2026 (Invited talk).*
5. Mixture of Experts in Large-scale and Multimodal Models. *MIT reading group hosted by Prof. Priya Donti, Virtual, 2025 (Invited talk).*
4. Mixture of Experts in Large-scale and Multimodal Models. *Two Sigma PhD Fellowship Finalist Reception, Virtual, 2025 (Invited talk).*
3. Mixture of Experts in Large-scale and Multimodal Models. *Dartmouth Applied and Computational Mathematics Seminar, Virtual, 2025 (Invited talk).*
2. Demystifying Softmax Gating Function in Gaussian Mixture of Experts. *STATML@UT Reading Group, Austin, TX, 2024 (Invited talk).*
1. Demystifying Softmax Gating Function in Gaussian Mixture of Experts. *IFML Workshop on Generative AI (Student talks session), Austin, TX, 2023 (Invited talk).*

Teaching Experience

The University of Texas at Austin, Austin, TX, USA.
Teaching Assistant at the Department of Statistics and Data Sciences.

- SDS302F - Foundations of Data Analysis. Fall 2022
- SDS322E - Elements of Data Science. Spring 2023
- SDS320E - Elements of Statistics. Fall 2023
- SDS315 - Statistical Thinking. Spring 2025

Professional Services

Senior Program Committee/Area Chair

- the Association for the Advancement of Artificial Intelligence (AAAI) 2027.

Program Committee/Reviewer at

- the Electronic Journal of Statistics (EJS).
- the Journal of Machine Learning Research (JMLR).
- the IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI).
- the Transactions on Machine Learning Research (TMLR).
- the International Conference on Machine Learning (ICML) 2022-2026.
- the Conference on Neural Information Processing Systems (NeurIPS) 2022-2026.
- the International Conference on Artificial Intelligence and Statistics (AISTATS) 2022-2027.
- the International Conference on Learning Representations (ICLR) 2024-2027.
- the Association for the Advancement of Artificial Intelligence (AAAI) 2025-2026.

Co-organizer of the Statistical Machine Learning seminar at UT Austin (STATML@UT).

Professional Membership

- Regular Member of the Institute of Electrical and Electronics Engineers (IEEE).
- Student Member of the Institute of Mathematical Statistics (IMS).

Honors and Awards

- UT Austin Outstanding Graduate Research Fellowship. 2026
- UT Austin Professional Development Award. 2024, 2025
- ICML 2026 Silver Reviewer. 2026
- Two Sigma PhD Fellowship Finalist. 2025
- Top Reviewer at NeurIPS 2024. 2024
- AISTATS 2024 Registration Grant. 2024
- ICLR 2024 Travel Award. 2024
- NeurIPS 2023 Scholar Award. 2023

Mentorship

- Viet Nguyen, *Undergrad student, Hanoi University of Science and Technology*
- Minh Pham, *Undergrad student, Purdue University*
- Thinh Cao, *Undergrad student, Hanoi University of Science and Technology*
- Tan Dinh, *Undergrad student, University of Massachusetts at Amherst*

References

- **Prof. Nhat Ho.** Email: minhnhat@utexas.edu (Advisor).
- **Prof. Alessandro Rinaldo.** Email: alessandro.rinaldo@austin.utexas.edu (Advisor).
- **Prof. Abbas Khalili.** Email: abbas.khalili@mcgill.ca (Collaborator).